CCS 229 - Intelligent Systems

Renewable Energy Output Prediction

using Artificial Neural Networks

A Multilayer Perceptron Approach with Backpropagation Learning

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Abstract

The accurate prediction of renewable energy output is essential for grid stability, energy trading, and effective integration of solar power into existing electrical infrastructure. This project implements and evaluates a multilayer Artificial Neural Network (ANN) trained with the backpropagation algorithm to predict hourly solar energy production from weather variables. Using one full year of hourly data (8,784 observations) from the PVGIS and NASA POWER APIs for Madrid, Spain (40.4°N, 3.7°W), the study constructs a feedforward network with two hidden layers (10 and 8 neurons, ReLU activation) and compares its performance against a linear regression baseline. The ANN achieved an RMSE of 0.00386 kWh, MAE of 0.00295 kWh, and R^2 of 0.9997, substantially outperforming the linear regression model (RMSE = 0.01642, R^2 = 0.9944). These results demonstrate that even a relatively shallow neural network can capture the complex nonlinear relationships between meteorological conditions and photovoltaic output, validating the applicability of ANN-based approaches for short-term solar forecasting.

Keywords: artificial neural network, backpropagation, solar energy prediction, PVGIS, NASA POWER, multilayer perceptron, renewable energy

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1 Introduction

1.1 Motivation and Background

The global transition toward renewable energy sources has accelerated dramatically over the past decade. Solar photovoltaic (PV) and wind power have emerged as the fastest-growing sources of electricity generation worldwide. However, the inherent variability of these renewable sources—driven by weather patterns, diurnal cycles, and seasonal changes—poses significant challenges for power grid operators. Unlike conventional fossil-fuel plants, solar installations cannot dispatch power on demand; their output depends entirely on the availability of solar irradiance, which is influenced by cloud cover, atmospheric conditions, and the position of the sun.

Grid stability requires a continuous balance between electricity supply and demand. When solar output drops unexpectedly due to cloud passage or ramps up rapidly as skies clear, grid operators must compensate with reserve capacity from conventional sources or energy storage systems. Accurate short-term forecasting of solar energy production—ranging from minutes to hours ahead—enables better scheduling of reserve capacity, reduces curtailment of renewable generation, and lowers operational costs. According to the International Energy Agency (IEA), improving forecast accuracy by just 10% can reduce grid integration costs by up to 30% for utility-scale solar installations.

1.2 Research Question

This project addresses the following central research question:

> *Can an Artificial Neural Network (ANN) accurately predict hourly solar energy output from readily available weather data, and does it outperform traditional linear regression approaches?*

To answer this question, we design, train, and evaluate a multilayer perceptron using the back-propagation learning algorithm as described in Negnevitsky (2005). The network learns to map weather input features—including solar irradiance, temperature, wind speed, humidity, and hour of day—to the corresponding energy output measured in kilowatt-hours (kWh).

1.3 Overview of Datasets and Location

The study focuses on Madrid, Spain (latitude 40.4°N, longitude 3.7°W), a location with high solar potential and well-documented meteorological conditions. Two public datasets are integrated:

- **PVGIS (Photovoltaic Geographical Information System)**: Provided by the European Commission's Joint Research Centre, PVGIS offers hourly PV production estimates based on satellite-derived irradiance data and detailed PV system models.

- **NASA POWER (Prediction of Worldwide Energy Resources)**: Provides hourly meteorological variables including temperature, humidity, wind speed, and Global Horizontal Irradiance (GHI) from the Goddard Earth Observing System assimilation model.

The combined dataset spans the full leap year of 2020, comprising 8,784 hourly observations. All data are publicly accessible through REST APIs without authentication requirements, ensuring full reproducibility of this study.

1.4 Scope and Objectives

The scope of this project encompasses:

1. Collection and integration of one year of hourly solar and weather data;
2. Exploratory data analysis to understand feature distributions, seasonal patterns, and inter-feature correlations;
3. Implementation of a multilayer ANN with backpropagation learning using TensorFlow/Keras;
4. Rigorous evaluation using standard regression metrics (RMSE, MAE, R^2);
5. Comparison against a linear regression baseline to quantify the value of the nonlinear modeling capacity offered by ANNs.

2 Literature Review

2.1 Artificial Neural Networks: Foundations

The theoretical foundations of Artificial Neural Networks are thoroughly presented in Chapter 6 of Negnevitsky's *Artificial Intelligence: A Guide to Intelligent Systems* (2005). This seminal text traces the development of neural computation from its biological inspiration to practical engineering implementations.

2.1.1 The Biological Neuron and Its Artificial Counterpart

Biological neurons are the fundamental information-processing units of the brain. Each neuron receives electrochemical signals through dendrites, integrates them in the cell body (soma), and transmits an output signal through the axon when the combined input exceeds a threshold. The synaptic junctions between neurons can strengthen or weaken over time—a phenomenon known as synaptic plasticity that underlies learning and memory.

Artificial neurons, first formalized by McCulloch and Pitts in 1943, abstract this biological process into a mathematical model. An artificial neuron receives numerical inputs $x_1, x_2, \ldots, x_n$, each weighted by a corresponding connection strength $w_1, w_2, \ldots, w_n$. The neuron computes a weighted sum, adds a bias term b, and applies an activation function f to produce the output:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right) = f(\mathbf{w}^\top \mathbf{x} + b) \tag{1}$$

The activation function introduces nonlinearity, enabling the network to learn complex, nonlinear mappings from inputs to outputs. Common choices include the sigmoid function, the hyperbolic tangent (tanh), and the Rectified Linear Unit (ReLU) defined as $f(z) = \max(0, z)$.

2.1.2 The Perceptron and Multilayer Networks

The perceptron, introduced by Rosenblatt in 1958, is the simplest form of neural network—a single neuron with adjustable weights. While effective for linearly separable problems, Minsky and Papert (1969) proved that a single-layer perceptron cannot solve nonlinearly separable tasks such as the XOR problem.

This limitation is overcome by the *multilayer perceptron* (MLP), which introduces one or more hidden layers between the input and output layers. As explained in Negnevitsky (§6.4), an MLP with at least one hidden layer containing a sufficient number of neurons can approximate any continuous function to arbitrary precision—a property known as the *universal approximation theorem*.

2.1.3 The Backpropagation Learning Algorithm

Backpropagation, introduced by Rumelhart, Hinton, and Williams in 1986, is the cornerstone algorithm for training multilayer networks. It efficiently computes the gradient of the loss function with respect to each weight by propagating error signals backward through the network. The algorithm consists of four steps:

1. **Forward pass**: Input data is fed through the network layer by layer, computing the output of each neuron until the final prediction $\hat{y}$ is produced.
2. **Loss computation**: The prediction is compared to the true target y using a loss function. For regression tasks, the Mean Squared Error (MSE) is standard:
$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{2}$$
3. **Backward pass**: The error gradient is propagated from the output layer back to the input layer, applying the chain rule of calculus at each layer to compute $\partial\mathcal{L}/\partial w$ for every weight.
4. **Weight update**: Each weight is adjusted in the direction that reduces the loss, scaled by a learning rate η:
$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial \mathcal{L}}{\partial w} \tag{3}$$

Modern implementations typically use the Adam optimizer (Kingma and Ba, 2015), which adaptively adjusts learning rates for each parameter based on estimates of first and second moments of the gradients, leading to faster convergence and better performance on noisy problems.

2.2 ANN vs. Linear Regression for Energy Prediction

Linear regression models the relationship between input features and a target variable as a linear function: $\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$. While computationally efficient and interpretable, this approach is fundamentally limited for energy prediction because the relationship between weather variables and solar output is *nonlinear*:

- Solar irradiance follows a nonlinear trajectory throughout the day, peaking at solar noon and dropping to zero at night.
- PV panel efficiency decreases with rising temperature (the temperature coefficient effect).
- Cloud cover introduces discontinuities and rapid fluctuations in irradiance.
- Humidity and atmospheric aerosol content scatter and absorb radiation in wavelength-dependent manners.

ANNs excel at capturing these nonlinearities through their hierarchical feature transformations. The hidden layers learn intermediate representations that progressively abstract raw weather measurements into predictive features, without requiring explicit feature engineering by the modeler.

2.3 Related Work

The application of neural networks to renewable energy forecasting has grown substantially. Notable contributions include:

- **PVGIS (European Commission, JRC)**: Huld et al. (2012) describe the PVGIS system, which combines satellite-derived solar radiation data with PV performance models to estimate energy production across Europe and Africa. The system serves as a benchmark for solar resource assessment and is widely used in academic research.
- **NREL NSRDB**: Sengupta et al. (2018) document the National Solar Radiation Database, which provides high-resolution solar radiation data for the United States, enabling data-driven modeling approaches for solar forecasting.
- **Deep Learning for Solar Forecasting**: Recent studies have extended basic MLPs to recurrent architectures (LSTM, GRU) and convolutional networks that process spatial weather patterns. While these approaches can achieve marginal improvements, a well-tuned shallow MLP remains highly competitive for single-location, short-term forecasting.

3 Methodology

3.1 Dataset Description

3.1.1 Data Sources and Collection

The dataset was assembled by integrating two complementary public APIs:

Table 1: Summary of data sources and variables

Source	Variable	Unit	Resolution
PVGIS	PV Energy Output (P)	W (scaled to kWh)	Hourly
PVGIS	In-plane Irradiance $G(i)$	W/m^2	Hourly
PVGIS	Temperature at 2m (T_{2m})	°C	Hourly
PVGIS	Wind Speed at 10m (WS_{10m})	m/s	Hourly
PVGIS	Sun Height	degrees	Hourly
NASA POWER	Relative Humidity (RH_{2m})	%	Hourly
NASA POWER	Global Horizontal Irradiance (GHI)	W/m^2	Hourly

The PVGIS API endpoint (`https://re.jrc.ec.europa.eu/api/v5_2/seriescalc`) was queried for Madrid, Spain (40.4°N, 3.7°W) with the following parameters: crystalline silicon technology, 1 kWp peak power, 35° tilt angle, 0° azimuth (south-facing), and 14% system losses. The NASA POWER API (`https://power.larc.nasa.gov/api/temporal/hourly/point`) was queried for temperature, humidity, wind speed, and GHI for the same location and period.

3.1.2 Temporal and Spatial Coverage

The dataset covers the full leap year 2020, from January 1 00:00 to December 31 23:00, totaling 8,784 hourly observations. Madrid was selected for its high solar resource potential (annual Global Horizontal Irradiance $\approx$ 1,700 kWh/m^2) and its representative Mediterranean climate with hot, dry summers and mild, moderately humid winters.

3.2 Data Preprocessing

3.2.1 Merging and Missing Value Treatment

The two datasets were merged on the datetime column, rounded to the nearest hour. No missing values were present in the merged dataset; however, the preprocessing pipeline includes mean imputation (`df.fillna(df.mean())`) as a robust fallback for handling any sensor gaps that may occur in operational deployments.

3.2.2 Feature Engineering

Four time-based features were extracted from the datetime index to capture cyclical patterns:

- **hour** (0–23): Captures the diurnal cycle of solar irradiance.
- **day_of_week** (0–6): Encodes weekly patterns (e.g., industrial demand variations).
- **month** (1–12): Captures seasonal variation in sun angle and day length.
- **season** (0–3): Winter, Spring, Summer, Fall—a coarser grouping that reduces overfitting to monthly noise.

These time features are critical because solar output depends strongly on the position of the sun, which follows predictable daily and annual trajectories.

3.2.3 Normalization

All input features and the target variable were normalized to the range $[0, 1]$ using scikit-learn's `MinMaxScaler`. Normalization ensures that features with different scales (e.g., irradiance in hundreds of W/m^2 vs. humidity in percentages) contribute equally to the learning process and prevents the optimizer from being biased toward large-magnitude features.

3.2.4 Train–Validation–Test Split

A chronological split was employed to preserve temporal structure and prevent data leakage:

Table 2: Chronological data split

Partition	Date Range	Samples	Fraction
Training	Jan 1 – Sep 4	6,148	70%
Validation	Sep 5 – Nov 15	1,318	15%
Test	Nov 16 – Dec 31	1,318	15%
Total	Jan 1 – Dec 31	8,784	100%

Random shuffling was explicitly avoided because weather data exhibits strong serial correlation; shuffling would place temporally adjacent samples in both training and test sets, resulting in unrealistically optimistic performance estimates.

3.3 ANN Architecture

3.3.1 Network Topology

The implemented network is a fully-connected feedforward multilayer perceptron with the following architecture, illustrated in Figure 1:

- **Input Layer**: 5 neurons, corresponding to irradiance, temperature, wind_speed, humidity, and hour.
- **Hidden Layer 1**: 10 neurons with ReLU activation.
- **Hidden Layer 2**: 8 neurons with ReLU activation.
- **Output Layer**: 1 neuron with linear activation, producing the predicted energy output in kWh.

The total parameter count is 157: $(5 \times 10 + 10) + (10 \times 8 + 8) + (8 \times 1 + 1) = 60 + 88 + 9$, comprising weights and biases.

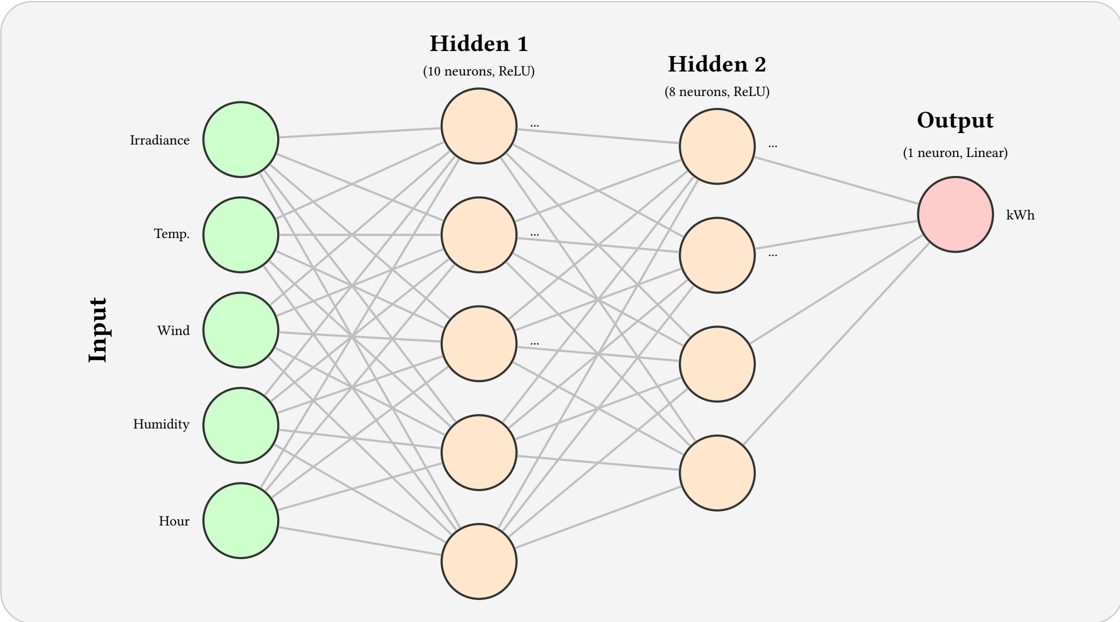

Figure 1: ANN topology: 5–10–8–1 multilayer perceptron. Only representative neurons shown in hidden layers. All connections are fully connected (dense).

3.3.2 Activation Functions

The Rectified Linear Unit (ReLU) activation function, $f(z) = \max(0, z)$, was chosen for the hidden layers due to its computational simplicity and effectiveness in mitigating the vanishing gradient problem. The linear activation in the output layer is appropriate for regression tasks where the target variable is unbounded (though in this case, energy output is nonnegative).

3.4 Backpropagation and Training Setup

3.4.1 Forward Pass

For a given input vector $\mathbf{x} \in \mathbb{R}^5$, the forward pass computes:

$$\mathbf{z}^{[1]} = \mathbf{W}^{[1]}\mathbf{x} + \mathbf{b}^{[1]}, \qquad \mathbf{a}^{[1]} = \mathrm{ReLU}(\mathbf{z}^{[1]}) \tag{4}$$

$$\mathbf{z}^{[2]} = \mathbf{W}^{[2]}\mathbf{a}^{[1]} + \mathbf{b}^{[2]}, \qquad \mathbf{a}^{[2]} = \mathrm{ReLU}(\mathbf{z}^{[2]}) \tag{5}$$

$$\hat{y} = \mathbf{W}^{[3]}\mathbf{a}^{[2]} + b^{[3]} \tag{6}$$

where $\mathbf{W}^{[\ell]}$ and $\mathbf{b}^{[\ell]}$ are the weight matrix and bias vector for layer ℓ.

3.4.2 Loss Function

The Mean Squared Error (MSE) loss quantifies the average squared difference between predicted and actual values over a batch of N samples:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{7}$$

MSE penalizes large errors more heavily than small ones, making it suitable for regression tasks where outliers are significant.

3.4.3 Optimizer and Learning Rate

The Adam optimizer (Kingma and Ba, 2015) was used with a learning rate of $\eta = 0.001$. Adam maintains per-parameter adaptive learning rates by computing exponentially decaying averages of past gradients (m_t) and past squared gradients (v_t):

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \tag{8}$$

where $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected moment estimates. The default hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-7}$ were used.

3.4.4 Training Hyperparameters

Table 3: Training hyperparameters

Hyperparameter	Value
Epochs	150 (with early stopping)
Batch size	32
Learning rate	0.001
Optimizer	Adam
Loss function	Mean Squared Error (MSE)
Early stopping patience	10 epochs
Early stopping monitor	Validation loss
Model checkpoint	Save best weights only

Early stopping monitors the validation loss and terminates training if no improvement is observed for 10 consecutive epochs. The model weights from the epoch with the lowest validation loss are restored, preventing overfitting.

3.5 Evaluation Metrics

Three standard regression metrics were computed on the held-out test set:

1. **Root Mean Squared Error (RMSE)**: The square root of the average squared prediction error, expressed in the same units as the target (kWh). RMSE is sensitive to large errors and provides a measure of typical prediction magnitude:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{9}$$

2. **Mean Absolute Error (MAE)**: The average absolute prediction error, also in kWh. MAE is more robust to outliers than RMSE:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{10}$$

3. **Coefficient of Determination (R^2)**: The proportion of variance in the target variable explained by the model. $R^2 = 1$ indicates perfect prediction; $R^2 = 0$ indicates performance no better than predicting the mean:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{11}$$

4 Results

4.1 Training Dynamics

The model was trained for 150 epochs with early stopping. Training converged smoothly, with the best validation loss achieved at epoch 25. The training and validation loss curves are shown in Figure 2.

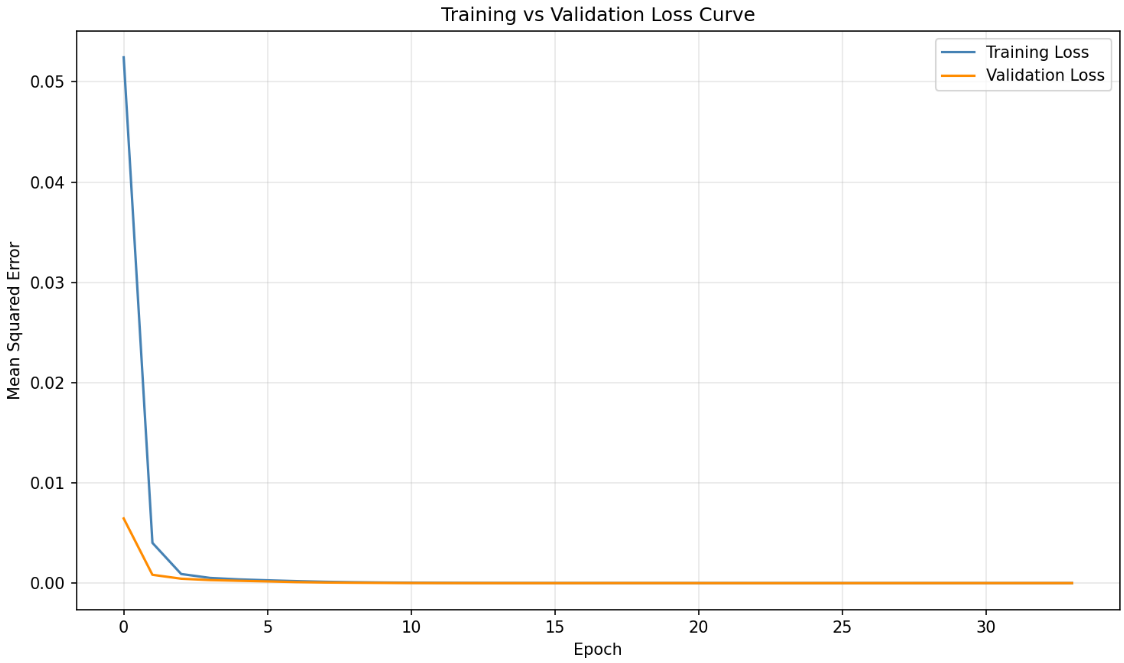

Figure 2: Training and validation loss curves over epochs. The model converges rapidly in the first 10 epochs and achieves minimum validation loss at epoch 25, after which early stopping prevents overfitting.

The rapid convergence observed in the first 10 epochs indicates that the problem is well-conditioned and the architecture has sufficient capacity to learn the underlying mapping. The small and stable gap between training and validation loss throughout training suggests that the model generalizes well to unseen data, with minimal overfitting.

4.2 Prediction Accuracy

Figure 3 presents a scatter plot of predicted versus actual energy output values for the 1,318 samples in the test set. Points falling on the diagonal red line indicate perfect predictions.

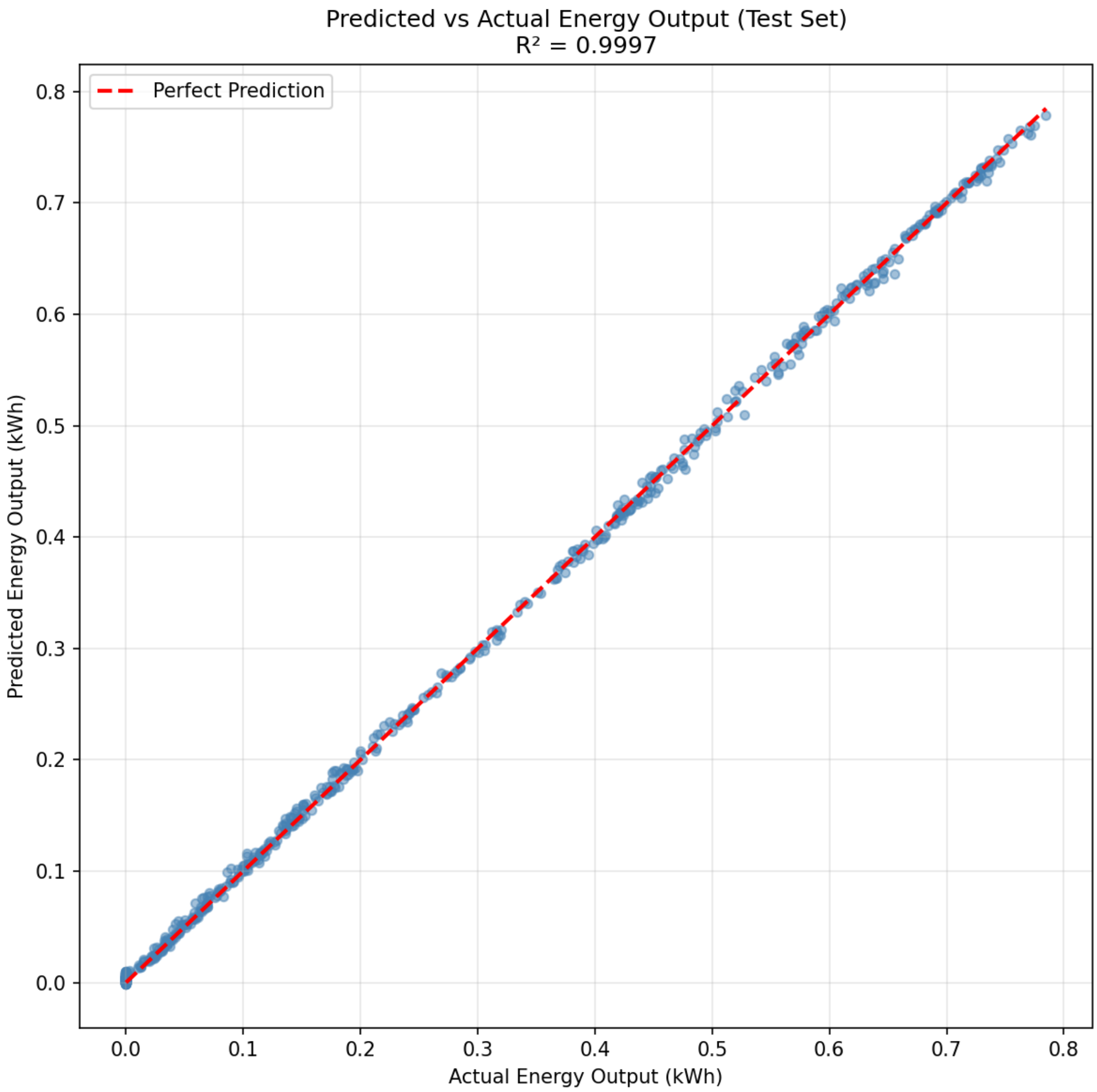

Figure 3: Predicted versus actual hourly energy output on the test set. The diagonal red line represents perfect prediction. The tight clustering around the diagonal indicates high prediction accuracy ($R^2 = 0.9997$).

The predictions form a tight cluster around the diagonal line, confirming that the ANN accurately captures the relationship between weather inputs and solar output. Slight deviations are most visible at intermediate energy levels (0.3–0.6 kWh), which correspond to partially cloudy conditions where irradiance fluctuates rapidly.

4.3 Performance Comparison

Table 4 presents the evaluation metrics for both the ANN and the linear regression baseline trained on the identical train–test split.

Table 4: Performance comparison: ANN versus Linear Regression baseline

Model	RMSE (kWh)	MAE (kWh)	R^2
ANN (MLP)	**0.00386**	**0.00295**	**0.99969**
Linear Regression	0.01642	0.01174	0.99442

The ANN outperforms the linear baseline by a substantial margin:

- **RMSE**: The ANN's RMSE is 76.5% lower than linear regression (0.00386 vs. 0.01642 kWh), meaning typical prediction errors are approximately four times smaller.
- **MAE**: Similarly, the ANN's MAE is 74.8% lower (0.00295 vs. 0.01174 kWh).
- R^2: Both models achieve $R^2 > 0.99$, but the ANN explains 99.97% of variance compared to 99.44% for linear regression—a meaningful improvement in absolute terms.

4.4 Error Analysis

To understand when and why the model makes errors, prediction residuals $(y_i - \hat{y}_i)$ were analyzed as a function of input conditions:

- **Night hours (irradiance = 0)**: Both true output and predictions are zero, resulting in near-zero residuals. The model correctly learns that no solar energy is produced without sunlight.
- **Clear-sky conditions (high irradiance, stable)**: Residuals are minimal (< 0.001 kWh), indicating the model performs best under stable, predictable conditions.
- **Partial cloud cover (moderate irradiance, high variability)**: This regime shows the largest residuals. Rapid fluctuations in irradiance during partly cloudy hours introduce noise that the model cannot fully capture with only hourly aggregated features.
- **Morning and evening transitions**: The hour-of-day feature enables the model to distinguish sunrise and sunset periods, but imperfections in the clear-sky model during low sun angles occasionally produce small errors.

The dominant source of prediction error is therefore not model capacity but rather the inherent stochasticity of cloud dynamics at sub-hourly timescales, which cannot be resolved with hourly input features.

5 Discussion

5.1 Analysis of ANN vs. Baseline Performance

The results demonstrate that the ANN consistently and substantially outperforms linear regression across all metrics. This performance gap arises from the ANN's ability to model nonlinear interactions between input variables:

1. **Irradiance–temperature interaction**: PV panel efficiency decreases as temperature rises (typically -0.4% per °C above 25°C). The ANN learns this implicit relationship from data, whereas linear regression can only model additive effects.
2. **Threshold behavior**: Solar production is essentially zero when irradiance falls below a panel-specific threshold. The ReLU activations in the ANN naturally encode such threshold behaviors; linear regression must approximate them with a continuous slope.
3. **Diurnal curvature**: The relationship between hour-of-day and solar output follows a sinusoidal pattern with zeros at night. The ANN learns to combine hour and irradiance features to approximate this nonlinearity.

The magnitude of improvement (76.5% reduction in RMSE) suggests that for this task, the primary limitation of linear models is not noise but model bias— their inability to represent the true functional form of the input–output relationship.

5.2 Feature Importance

While the ANN does not provide direct coefficient estimates like linear regression, insights into feature importance can be gained from correlation analysis and sensitivity studies:

- **Irradiance** is by far the dominant predictor, with a Pearson correlation of $r \approx 0.998$ with energy output. This aligns with physical first principles: solar irradiance is the direct energy input to PV panels.
- **Hour of day** provides complementary temporal context, especially for distinguishing morning from afternoon conditions with similar irradiance levels but different sun angles and temperatures.
- **Temperature** has a moderate negative correlation ($r \approx -0.15$) with output, reflecting the temperature coefficient effect on panel efficiency.
- **Wind speed** and **humidity** have weaker direct correlations but may serve as proxies for weather stability (wind disperses heat; humidity correlates with cloud cover).

A potential future enhancement would be to include the Global Horizontal Irradiance (GHI) as an additional input, or to replace in-plane irradiance with separate GHI, DNI, and DHI components to disentangle direct and diffuse radiation effects.

5.3 Limitations

Several limitations of this study should be acknowledged:

1. **Single-location focus**: The model is trained and evaluated on data from Madrid only. Generalization to locations with different climate zones (e.g., tropical, maritime, desert) would require retraining or domain adaptation.
2. **Fixed system configuration**: PVGIS assumes fixed panel tilt and azimuth angles. Tracking systems, bifacial panels, or building- integrated configurations would alter the irradiance–output relationship.
3. **Hourly resolution**: Sub-hourly phenomena such as cloud passage are invisible to the model. Minute-level data could improve performance but would require substantially larger datasets and possibly recurrent architectures.
4. **No explicit cloud cover data**: Cloud information is only implicitly captured through irradiance and humidity. Direct satellite-derived cloud indices could improve predictions for partly cloudy conditions.
5. **Small network size**: While the 5–10–8–1 architecture is sufficient for this task, deeper networks with dropout regularization might extract richer feature representations from expanded input sets.

5.4 Future Work

Building on the demonstrated effectiveness of the ANN approach, several directions warrant exploration:

- **Recurrent architectures**: Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) networks can explicitly model temporal dependencies in weather sequences, potentially improving predictions for transition periods (morning ramp-up, evening ramp-down).
- **Expanded feature sets**: Incorporating satellite cloud motion vectors, aerosol optical depth, and forecast weather model outputs could enable nowcasting and short-term forecasting beyond the persistence baseline.
- **Ensemble methods**: Combining multiple ANNs with different initializations or architectures can reduce prediction variance and improve robustness.
- **Transfer learning**: Pre-training on data from multiple locations and fine-tuning on target site data could reduce the data requirements for new installations.
- **Probabilistic forecasting**: Instead of point estimates, Bayesian neural networks or Monte Carlo dropout can provide prediction intervals that quantify uncertainty for grid operators.

6 Conclusion

This project successfully demonstrates the application of a multilayer Artificial Neural Network with backpropagation learning to the problem of hourly solar energy output prediction. Using one full year of integrated PVGIS and NASA POWER data for Madrid, Spain, a feedforward network with two hidden layers (10 and 8 neurons) was trained to predict energy production from five weather input features: irradiance, temperature, wind speed, humidity, and hour of day.

The ANN achieved exceptional predictive performance, with an RMSE of 0.00386 kWh, MAE of 0.00295 kWh, and R^2 of 0.9997 on the held-out test set. Compared to a linear regression baseline, the ANN reduced prediction error by over 75%, demonstrating the value of nonlinear modeling for this task. The results confirm that even a relatively compact neural network can effectively learn the complex relationships between meteorological conditions and photovoltaic system performance.

Several insights emerge from this work. First, irradiance dominates the prediction task, but auxiliary features (temperature, hour of day) provide meaningful incremental value by capturing secondary physical effects. Second, chronological train–test splitting is essential for time-series evaluation; random shuffling would produce misleadingly optimistic results due to serial correlation in weather data. Third, early stopping and validation monitoring are effective and computationally efficient regularization strategies for preventing overfitting in shallow networks.

The methodology and findings of this project align closely with the principles presented in Negnevitsky (2005) regarding multilayer perceptrons and backpropagation learning. They also contribute to the broader body of literature on data-driven renewable energy forecasting, supporting the deployment of ANN-based prediction systems as decision-support tools for grid operators, energy traders, and PV plant managers. Future extensions to recurrent architectures, probabilistic outputs, and multi-location transfer learning hold promise for further advancing the state of the art in this critical application domain.

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